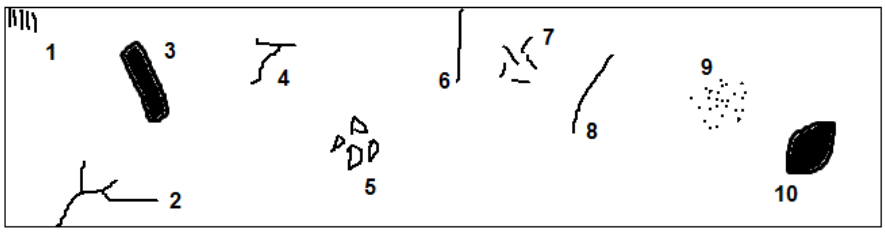
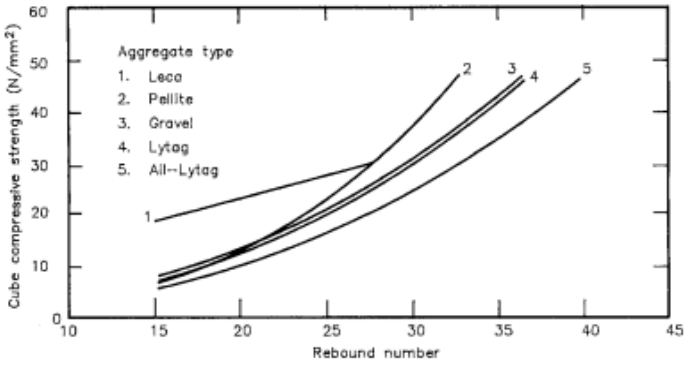
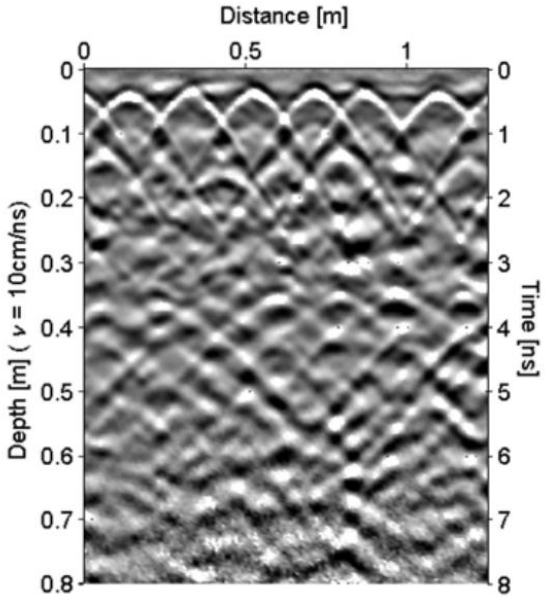
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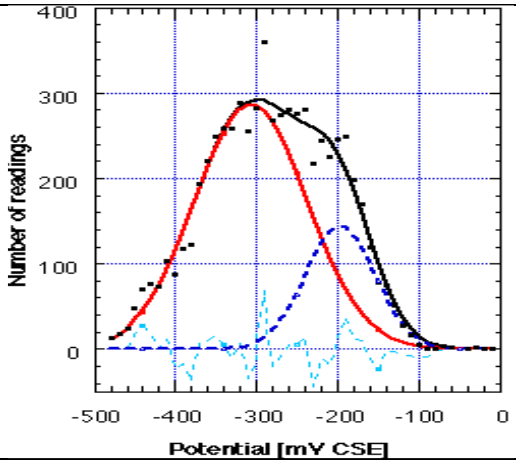
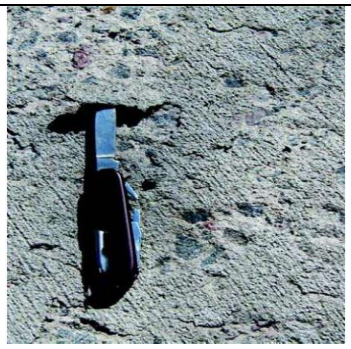
DEPARTMENT OF CIVIL ENGINEERING			
END SEMESTER EXAMINATION - MAKEUP		**2024**	
Level	**B.Tech**	**Curriculum**	**2022**
Program Name	**Civil Engineering**	**Course Type**	**PE - VII**
Course Name	**Non-destructive Testing (NDT) of Concrete Structures**	**Course Code**	**CIE 4080**
Semester	**VII**	**Max Marks**	**50**
Date		**Duration**	**3 hours**

Instructions:

- **Answer ALL the questions.**
- **Data not provided may be suitably assumed**

Q. No.	Question	Marks	NBA		BL
			CO	PO	
1A	Explain the role of non-destructive testing in assuring the safety of concrete structures.	5	1	1,2,6, 12	3
1B	Explain the applications of X-rays in inspecting concrete and mention its limitations.	3	1	1,2,6, 12	3
1C	Explain the characteristics of magnetic particles used in magnetic particle inspection (LPI).	2	2	5,6,9, 12	3
2A	Illustrate with a neat sketch, the working of the Windsor probe technique for concrete strength determination.	5	2	5,6,9, 12	3
2B	Suggest and explain a suitable magnetic particle inspection (MPI) method for inspecting the steel column containing defects near the surface.	3	2	5,6,9, 12	3
2C	Illustrate the process of determining the surface defect in metals using eddy current.	2	3	5,8,12	3
3A	With a neat sketch, explain the procedure of direct and semi-direct methods of inspection of concrete using ultrasound.	5	3	5,8,12	3
3B	Illustrate the procedure of determining the reinforcement position in concrete with a radargram using the ground penetrating radar (GPR) technique.	3	3	5,8,12	3
3C	Explain the role of couplant used in ultrasound inspection (UI) technique.	2	3	5,8,12	3
4A	Draw the typical waveforms produced on A scan display for the defects shown below that are inspected using ultrasound.	5	5	1,2,3, 4,12	4

					
4B	The concrete specimens comprising different aggregate types were tested using a rebound hammer. The result of the test is presented in the figure as shown below. Analyse of the effect of perlite, gravel and lytag on rebound number and strength of the concrete. 	**3**	5	**1,2,3,4 ,12**	**4**
4C	Analyse the features of the radargram shown below for the reinforced concrete specimen. 	**2**	5	**1,2,3,4 ,12**	**4**
5A	Analyse the corrosion potential graph, as shown above, obtained after conducting the half-cell potential test on reinforced concrete	**5**	5	**1,2,3,4 ,12**	**4**

					
5B	Analyse the defects in concrete structure as shown the below figures.  **(a)**  **(b)**  **(c)**	**3**	5	**1,2,3,4 ,12**	**3**
5C	Discuss the effect of embedded reinforcement in concrete on ultrasound's pulse velocity and concrete's compressive strength.	**2**	5	**1,2,3,4 ,12**	**3**

MARK DISTRIBUTION (CO-wise & BTL-wise)

CO / CLO	Total Marks
1	***8***
2	***2***
3	***18***
4	***0***
5	***22***

BL	Total Marks
2	**0**
3	**35**
4	**15**
5	**0**